1 Generated reference tables

The tables in this appendix are generated from the source tree by `scripts/gen_report_tables.py`. They are not maintained by hand: a renamed flag, a new cycle-slip detector, or a changed state-key delimiter appears here on the next run of the generator. *Known gap:* the feature-flag table de-duplicates on the C++ member name. A section master switch named `enable` in more than one config sub-struct therefore appears once, and several `network.*` master switches are absent. The table is indicative. The loaders `GnssConfigLoad.cpp` and `GnssConfigLoadNetwork.cpp` (`src/core/config/`) are authoritative, and the verdict on every key the network loader reads (armed, recommended, kept off, unvalidated) is recorded in `docs/CONFIG_promotion_ledger.json`. The fragments under `generated/` are git-ignored and rebuilt on demand (`build.ps1 -Regen` / `build.sh --regen`). `gen_report_tables.py --check` exits non-zero when a fragment changed; it rewrites the fragments before comparing, so after a failed check the regenerated tables are already in place.

1.1 Processing modes

Table 1: `ProcessingMode` – the estimation algorithm selector.

Value	Meaning
`PPP`	Precise Point Positioning (single receiver, external products).
`PPP_RTK`	PPP with RTK enhancements (single receiver, uncombined).
`BaselineSingle`	Double-differenced single baseline: one rover + one reference station.
`BaselineMulti`	Double-differenced multi-receiver network (star topology: N rovers against one reference).
`PPP_RTK_Network`	Minimum-constraint S-system network producer: joint multi-station estimator emitting satellite clock + code/phase bias (+ atmosphere) under a declared S-basis.
`PPP_RTK_User`	PPP-RTK consumer of network PPP_RTK_Network products (post-processing closed-loop counterpart): uncombined PPP-RTK + network SSR bias/atmosphere consume + S-basis cross-check.

1.2 Execution mode and data source

Table 2: `AppMode` – top-level post-processing vs. real-time dispatch.

Value	Meaning
`PostProcessing`	Batch processing from RINEX / product files.
`RealTime`	Streaming / live processing from RTCM data.

Table 3: `DataSource` – bundled fixtures vs. auto-download.

Value	Meaning
`Internal`	Use files under internal_data_root only.
`Production`	Auto-download from CDDIS into data_root.

1.3 Differencing regime

Table 4: `DifferencingMode` – the measurement differencing regime.

Value	Meaning
`None`	Raw observables (PPP / PPP-RTK without SD).
`BetweenSatellite`	Per-constellation single difference vs. pivot (PPP-RTK SD).
`DoubleDifference`	Between-receiver + between-satellite (baseline processing).

1.4 Config-to-differencing mapping

The differencing regime is resolved once per run by the factory below (verbatim from `MeasurementBuilder.cpp`):

```
DifferencingMode differencingModeFor(const GnssConfig& config) {
    // The network producer runs on undifferenced, uncombined rows and estimates the
    // satellite-side bias lumps explicitly, so it never applies receiver-side BSSD,
    // even when a between_satellite_sd toggle is present in its config.
    if (config.getProcessingMode() == ProcessingMode::PPP_RTK_Network)
                                        return DifferencingMode::None;
    if (config.isBaselineMode())        return DifferencingMode::DoubleDifference;
    if (config.enableBetweenSatelliteSd()) return DifferencingMode::BetweenSatellite;
    return DifferencingMode::None;
}
```

1.5 Feature toggles

These are the loop- and state-vector-level switches.

Table 5: Top-level boolean feature toggles, parsed from the `jval<bool>` calls in `GnssConfigLoad.cpp` and `GnssConfigLoadNetwork.cpp`. The legacy alias is the pre-refactor flat key still accepted as a fallback.

JSON path	Default	Legacy alias
`estimation.accumulate_phase_windup`	off	`accumulate_phase_windup`
`estimation.adaptive_iono_process_noise.trace`	off	`adaptive_iono_q_trace`
`estimation.aiding.baro`	off	`aiding_baro`

JSON path	Default	Legacy alias
estimation.aiding.nhc	off	aiding_nhc
estimation.aiding.vision	off	aiding_vision
estimation.aiding.wheel	off	aiding_wheel
estimation.aiding.zupt	off	aiding_zupt
estimation.ambiguity_resolution.enable_partial_ar	off	ar_enable_partial_ar
estimation.ambiguity_resolution.nl_iono_free_row	off	ar_nl_iono_free_row
estimation.ambiguity_resolution.wl_only	off	ar_wl_only
estimation.ambiguity_resolution.wl_only_ffr	on	ar_wl_only_ffr
estimation.between_satellite_sd.enable_glonass	off	enable_glonass_bssd
estimation.between_satellite_sd.enabled	off	enable_between_satellite_sd
estimation.cycle_slip.ewl	off	cycle_slip_ewl
estimation.cycle_slip.gf	off	cycle_slip_gf
estimation.cycle_slip.graphic	off	cycle_slip_graphic
estimation.cycle_slip.graphic_trend_centred	off	cycle_slip_graphic_trend_centred
estimation.cycle_slip.mw	on	cycle_slip_mw
estimation.cycle_slip.turbo_edit	off	cycle_slip_turbo_edit
estimation.cycle_slip.turbo_edit_bridge_lli	off	cycle_slip_turbo_edit_bridge_lli
estimation.cycle_slip.turbo_edit_mw_only_veto	on	cycle_slip_turbo_edit_mw_only_veto
estimation.defer_until_corrections_ready.drop_broadcast_grade	off	defer_until_corrections_ready_drop_broadcast_grade
estimation.defer_until_corrections_ready.require_all_above_mask	off	defer_until_corrections_ready_require_all_above_mask
estimation.enable_cio_frame	off	enable_cio_frame
estimation.enable_data_snooping	off	enable_data_snooping
estimation.enable_full_r_kernel	off	enable_full_r_kernel
estimation.enable_huber_innovation	off	enable_huber_innovation
estimation.enable_huber_r_inflation	on	enable_huber_r_inflation
estimation.enable_if_slip_detector	off	enable_if_slip_detector
estimation.enable_iono_rate	off	enable_iono_rate
estimation.enable_subcycle_slip_suppression	on	enable_subcycle_slip_suppression
estimation.enable_sun_frame_correction	off	enable_sun_frame_correction
estimation.enable_variance_factor	off	enable_variance_factor
estimation.estimate_ifb	off	estimate_ifb
estimation.estimate_phase_clock	off	estimate_phase_clock

JSON path	Default	Legacy alias
estimation.estimate_sat_biases	off	estimate_sat_biases
estimation.estimate_sat_clocks	off	estimate_sat_clocks
estimation.estimate_sat_phase_biases	off	estimate_sat_phase_biases
estimation.estimate_tropo_gradients	off	estimate_tropo_gradients
estimation.estimate_velocity	off	estimate_velocity
estimation.global_vtec_sh.rep_error_d5_keep	off	global_vtec_sh_rep_error_d5_keep
estimation.global_vtec_sh.rep_error_demean	off	global_vtec_sh_rep_error_demean
estimation.global_vtec_sh.sun_fixed	on	global_vtec_sh_sun_fixed
estimation.glonass_ifb.enable	off	enable_glonass_ifb
estimation.lock_signals_to_clock_ref	off	lock_signals_to_clock_ref
estimation.phase_windup_gap_bridge	off	phase_windup_gap_bridge
estimation.promote_sat_code_bias	off	promote_sat_code_bias
estimation.require_bia_for_signals	off	require_bia_for_signals
estimation.residual_history.enabled	off	residual_history_enabled
estimation.trace_windup_census	off	trace_windup_census
network.adm_revet	off	adm_revet
network.adm_revet_census	on	adm_revet_census
network.adm_revet_hold	off	adm_revet_hold
network.adm_screen	off	adm_screen
network.adm_trace	off	adm_trace
network.amb_float_declared_handover	off	amb_float_declared_handover
network.amb_float_drop_decouple_always	off	amb_float_drop_decouple_always
network.ambiguity_resolution.block_search	off	network_ar_block_search
network.ambiguity_resolution.cycle_basis	off	network_ar_cycle_basis
network.ambiguity_resolution.cycle_basis_sticky	on	network_ar_cycle_basis_sticky
network.ambiguity_resolution.decorrelated_partial	off	network_ar_decorrelated_partial
network.ambiguity_resolution.enable_partial_ar	on	network_ar_enable_partial_ar
network.ambiguity_resolution.trace_lattice	off	network_ar_trace_lattice
network.ambiguity_resolution.tree_basis	off	network_ar_tree_basis
network.block_baarda	off	block_baarda
network.block_baarda_corroborate	off	block_baarda_corroborate
network.block_baarda_estimable	off	block_baarda_estimable
network.canonical_pair_admission	off	canonical_pair_admission
network.carry_aware_covariance	off	network_carry_aware_covariance

JSON path	Default	Legacy alias
network.carry_conditional_coupling	off	carry_conditional_coupling
network.carry_coupling_bias_clone	off	carry_coupling_bias_clone
network.carry_coupling_clock_clone	off	carry_coupling_clock_clone
network.carry_coupling_departed_clone	off	carry_coupling_departed_clone
network.carry_coupling_exact_cross	off	carry_coupling_exact_cross
network.carry_coupling_exact_update	off	carry_coupling_exact_update
network.carry_coupling_gauge_project	off	carry_coupling_gauge_project
network.carry_coupling_reinit_sources	off	carry_coupling_reinit_sources
network.carry_coupling_shared_history	off	carry_coupling_shared_history
network.carry_coupling_time_update	off	carry_coupling_time_update
network.carry_invariant	off	network_carry_invariant
network.carry_invariant_emit_rebase	on	network_carry_invariant_emit_rebase
network.carry_invariant_innov_calibrate	off	network_carry_invariant_innov_calibrate
network.carry_invariant_innov_carried_only	off	network_carry_invariant_innov_carried_only
network.carry_invariant_innov_corroborate	off	network_carry_invariant_innov_corroborate
network.carry_invariant_innov_gate	off	network_carry_invariant_innov_gate
network.carry_invariant_innov_retap_only	off	network_carry_invariant_innov_retap_only
network.carry_invariant_pd_safe	off	network_carry_invariant_pd_safe
network.carry_iono_in_block	off	carry_iono_in_block
network.carry_rebase_complete	off	network_carry_rebase_complete
network.carry_rx_clock_in_block	off	carry_rx_clock_in_block
network.carry_rx_clock_isb	on	carry_rx_clock_isb
network.carry_rx_code_bias	off	carry_rx_code_bias
network.carry_rx_code_bias_in_block	off	carry_rx_code_bias_in_block
network.carry_vce	off	carry_vce
network.carry_vce_apply	off	carry_vce_apply
network.carry_zwd	off	carry_zwd
network.carry_zwd_in_block	off	carry_zwd_in_block
network.cb_admission	off	cb_admission
network.cb_excursion_reinit	off	cb_excursion_reinit
network.cb_excursion_reinit_all_sats	off	cb_excursion_reinit_all_sats
network.cb_kaula_prior	off	network_cb_kaula_prior

JSON path	Default	Legacy alias
network.clock_anchor_datum_keep	off	network_clock_anchor_datum_keep
network.clock_anchor_gauge_free	off	network_clock_anchor_gauge_free
network.clock_anchor_mark	on	network_clock_anchor_mark
network.clock_anchor_resolve	on	network_clock_anchor_resolve
network.clock_datum_broadcast_anchored	off	network_clock_datum_broadcast_anchored
network.clock_datum_sum_to_zero	off	network_clock_datum_sum_to_zero
network.clock_disc_jump_test	off	network_clock_disc_jump_test
network.clock_disc_on_set_change	off	network_clock_disc_on_set_change
network.clock_level_bound_raise	off	network_clock_level_bound_raise
network.clock_level_monitor	off	network_clock_level_monitor
network.clock_level_restore_raise	off	clock_level_restore_raise
network.clock_model	off	network_clock_model
network.clock_monitor	off	network_clock_monitor
network.clock_monitor_couple_raim	off	network_clock_monitor_couple_raim
network.clock_pivot_frame_q	off	clock_pivot_frame_q
network.clock_runaway_check	off	network_clock_runaway_check
network.clock_temporal_anchor	off	network_clock_temporal_anchor
network.code_bias_disc_jump_test	off	network_code_bias_disc_jump_test
network.code_datum_family_difference	off	network_code_datum_family_difference
network.code_datum_pair_continuity	off	network_code_datum_pair_continuity
network.code_datum_time_linked	off	network_code_datum_time_linked
network.cold_clock_seed	off	cold_clock_seed
network.congruence_screen	off	congruence_screen
network.consume_phase_bias_sigma	off	network_consume_phase_bias_sigma
network.consume_ura	off	network_consume_ura
network.ctu_switch_on_resume	off	ctu_switch_on_resume
network.cycle_slip.anchor_slip_handover	off	network_cycle_slip_anchor_slip_handover

JSON path	Default	Legacy alias
network.cycle_slip.common_band_ref_arc	off	network_cycle_slip_common_band_ref_arc
network.cycle_slip.common_band_scope	off	network_cycle_slip_common_band_scope
network.cycle_slip.datum_station_detect	off	network_cycle_slip_datum_station_detect
network.cycle_slip.gap_instrument	off	network_cycle_slip_gap_instrument
network.cycle_slip.gap_reentry_coresumed_share	off	network_cycle_slip_gap_reentry_coresumed_share
network.cycle_slip.gap_reentry_per_signal	off	network_cycle_slip_gap_reentry_per_signal
network.cycle_slip.gap_reentry_per_signal_unmeasurable	on	network_cycle_slip_gap_reentry_per_signal_unmeasurable
network.cycle_slip.gap_reentry_relevel	off	network_cycle_slip_gap_reentry_relevel
network.cycle_slip.gap_reentry_relevel_lli	on	network_cycle_slip_gap_reentry_relevel_lli
network.cycle_slip.gap_reentry_relevel_unmeasurable	off	network_cycle_slip_gap_reentry_relevel_unmeasurable
network.cycle_slip.gf_arc_jump	off	network_cycle_slip_gf_arc_jump
network.cycle_slip.if_phase_td	off	network_cycle_slip_if_phase_td
network.cycle_slip.if_phase_td_ref_arm_all	off	network_cycle_slip_if_phase_td_ref_arm_all
network.cycle_slip.if_phase_td_ref_elev_norm	off	network_cycle_slip_if_phase_td_ref_elev_norm
network.cycle_slip.if_phase_td_ref_gate	off	network_cycle_slip_if_phase_td_ref_gate
network.cycle_slip.if_phase_td_ref_warmup_abstain	off	network_cycle_slip_if_phase_td_ref_warmup_abstain
network.cycle_slip.if_phase_td_reseed_on_flag	off	network_cycle_slip_if_phase_td_reseed_on_flag
network.cycle_slip.mw_detect	off	network_cycle_slip_mw_detect
network.cycle_slip.mw_evidence_refloat	off	network_cycle_slip_mw_evidence_refloat
network.cycle_slip.mw_lli_corroborate	off	network_cycle_slip_mw_lli_corroborate
network.cycle_slip.mw_reflag_skip	off	network_cycle_slip_mw_reflag_skip

JSON path	Default	Legacy alias
network.cycle_slip.oracle_relevel	off	network_cycle_slip_oracle_relevel
network.cycle_slip.relevel_bridge_gf	off	network_cycle_slip_relevel_bridge_gf
network.cycle_slip.relevel_delta_bridge	off	network_cycle_slip_relevel_delta_bridge
network.cycle_slip.relevel_free_epoch	off	network_cycle_slip_relevel_free_epoch
network.cycle_slip.tracker_modeled_first	off	network_cycle_slip_tracker_modeled_first
network.cycle_slip.turbo_edit	off	network_cycle_slip_turbo_edit
network.cycle_slip.windup_restart_refloat	off	network_cycle_slip_windup_restart_refloat
network.datum_graph_census	off	datum_graph_census
network.emit_clock_disc	off	network_emit_clock_disc
network.emit_clock_frame_hold	off	network_emit_clock_frame_hold
network.emit_combined_sigma	off	network_emit_combined_sigma
network.emit_epoch_status_record	off	network_emit_epoch_status_record
network.emit_generation_id	off	network_emit_generation_id
network.emit_generation_time	off	network_emit_generation_time
network.emit_pb_hold	off	emit_pb_hold
network.emit_pb_hold_phase_clock	off	emit_pb_hold_phase_clock
network.emit_pb_hold_smoothed	off	emit_pb_hold_smoothed
network.emit_phase_bias_sigma	off	network_emit_phase_bias_sigma
network.emit_ura	off	network_emit_ura
network.estimate_glonass	off	estimate_glonass
network.estimate_glonass	off	estimate_glonass
network.estimate_iono_sh	off	estimate_iono_sh
network.estimate_rx_extra_band_code_delay	off	estimate_rx_extra_band_code_delay
network.estimate_sat_code_bias	off	estimate_sat_code_bias
network.exclude_unmodeled_antenna_stations	off	network_exclude_unmodeled_antenna_stations
network.gross_los_bound	off	gross_los_bound
network.gross_los_emit_exclude	off	gross_los_emit_exclude
network.gross_los_sat_standdown	off	gross_los_sat_standdown
network.gross_los_sat_wire_withhold	on	gross_los_sat_wire_withhold
network.gross_los_screen	off	gross_los_screen
network.gross_los_station_tier	off	gross_los_station_tier

JSON path	Default	Legacy alias
network.h_sigma0_prefit_census	off	network_h_sigma0_prefit_census
network.handover_stransform	off	network_handover_stransform
network.handover_stransform_before_rows	off	network_handover_stransform_before_rows
network.handover_stransform_dense	off	network_handover_stransform_dense
network.iono_supply_formal_sigma	off	iono_supply_formal_sigma
network.ledger_anchor	off	network_ledger_anchor
network.ledger_emit	off	network_ledger_emit
network.ledger_reentry	off	network_ledger_reentry
network.ledger_screens	off	network_ledger_screens
network.level_authority_csgf	off	network_level_authority_csgf
network.level_authority_los_dump	off	network_level_authority_los_dump
network.mark_refsat_reset_disc	off	network_mark_refsat_reset_disc
network.mdb	off	mdb
network.n3c_band_signal	off	network_n3c_band_signal
network.n3c_bia_pin_pair	off	network_n3c_bia_pin_pair
network.n3c_canonical_pair_traits	off	network_n3c_canonical_pair_traits
network.n3c_channel_family	off	network_n3c_channel_family
network.n3c_ledger_lifecycle	off	network_n3c_ledger_lifecycle
network.n3c_wire_per_signal	off	network_n3c_wire_per_signal
network.parallel_carry	off	parallel_carry
network.parallel_seam	off	parallel_seam
network.parallel_stations	on	parallel_stations
network.per_const_withhold	off	network_per_const_withhold
network.phase_anchor_carry_reref	off	network_phase_anchor_carry_reref
network.phase_colour_state	off	phase_colour_state
network.phase_datum_candidacy_satellite	on	network_phase_datum_candidacy_satellite
network.phase_datum_candidacy_usable	off	network_phase_datum_candidacy_usable
network.phase_datum_eq34	off	network_phase_datum_eq34
network.phase_datum_eq34_carry_shift	off	network_phase_datum_eq34_carry_shift
network.phase_datum_eq34_extend_only	off	network_phase_datum_eq34_extend_only

JSON path	Default	Legacy alias
network.phase_datum_eq34_longevity_election	off	network_phase_datum_eq34_longevity_election
network.phase_datum_eq34_pin_round	on	network_phase_datum_eq34_pin_round
network.phase_datum_eq34_seed_round	off	network_phase_datum_eq34_seed_round
network.phase_datum_eq34_seed_round_rehost_frame	off	network_phase_datum_eq34_seed_round_rehost_frame
network.phase_datum_eq34_slip_aware_election	off	network_phase_datum_eq34_slip_aware_election
network.phase_datum_gate_drop_erase	off	network_phase_datum_gate_drop_erase
network.phase_datum_handover_continuous	off	network_phase_datum_handover_continuous
network.phase_datum_held_basis	off	network_phase_datum_held_basis
network.phase_datum_held_cold_release	off	network_phase_datum_held_cold_release
network.phase_datum_held_mark_cold	off	network_phase_datum_held_mark_cold
network.phase_datum_prioritized	off	network_phase_datum_prioritized
network.phase_datum_sticky	on	network_phase_datum_sticky
network.phase_datum_time_linked	off	network_phase_datum_time_linked
network.phase_leveling	off	phase_leveling
network.phase_leveling_local_block	off	phase_leveling_local_block
network.phase_visible_monitor	off	phase_visible_monitor
network.pivot_vanish_return_signal	off	network_pivot_vanish_return_signal
network.prior_residual_census	off	prior_residual_census
network.prior_residual_corroborate	off	prior_residual_corroborate
network.prior_residual_global_sigma0	off	prior_residual_global_sigma0
network.product_flush_per_epoch	off	product_flush_per_epoch
network.raim	off	network_raim
network.raim_attribution	off	network_raim_attribution
network.raim_carry_invariant	off	network_raim_carry_invariant
network.raim_pool_data_rows_only	off	network_raim_pool_data_rows_only
network.raim_rate_hygiene	off	network_raim_rate_hygiene
network.raim_redundancy_normalize	off	network_raim_redundancy_normalize
network.raim_refloat_maturity_gate	off	network_raim_refloat_maturity_gate

JSON path	Default	Legacy alias
network.raim_station_factor	off	network_raim_station_factor
network.raim_surgical_station	on	network_raim_surgical_station
network.raim_we_code_iono	off	network_raim_we_code_iono
network.raim_withhold_free_erase	off	network_raim_withhold_free_erase
network.raim_within_epoch	off	network_raim_within_epoch
network.readmit_seed_code_bias	on	readmit_seed_code_bias
network.readmit_seed_mirrors	off	readmit_seed_mirrors
network.replay_allow_undeclared_wire	off	replay_allow_undeclared_wire
network.require_signal_integer_for_ar	off	network_require_signal_integer_for_ar
network.resume_border_restore	off	resume_border_restore
network.resume_coupling_restore	off	resume_coupling_restore
network.resume_datum_restore	off	resume_datum_restore
network.resume_emit_restore	off	resume_emit_restore
network.resume_frame_restore	off	resume_frame_restore
network.resume_pb_seam_mark	off	resume_pb_seam_mark
network.resume_rxcb_lin_restore	off	resume_rxcb_lin_restore
network.resume_signal_canon_restore	off	resume_signal_canon_restore
network.resume_station_restore	off	resume_station_restore
network.resume_station_restore_rxcb	on	resume_station_restore_rxcb
network.rhs_attrib_on_trip	off	rhs_attrib_on_trip
network.rhs_attrib_on_trip	off	rhs_attrib_on_trip
network.rhs_attrib_on_trip_prev	on	rhs_attrib_on_trip_prev
network.roster_tracker	off	network_roster_tracker
network.rx_code_bias_datum	off	rx_code_bias_datum
network.sat_gf_datum_time_linked	off	sat_gf_datum_time_linked
network.sat_gf_extra_band_free	off	sat_gf_extra_band_free
network.sat_isc_state	off	sat_isc_state
network.sbasis_complete	off	network_sbasis_complete
network.scoped_handover_response	off	network_scoped_handover_response
network.sd_ewl.fix_enable	off	network_sd_ewl_fix_enable
network.second_band_admission	off	second_band_admission
network.second_code_channel	off	second_code_channel
network.second_phase_channel	off	second_phase_channel
network.serialize_disc_counters	off	network_serialize_disc_counters
network.service_mode	off	network_service_mode
network.shared_carry_full_cov	off	shared_carry_full_cov

JSON path	Default	Legacy alias
network.signal_integrity.census	off	network_signal_integrity_census
network.signal_integrity.refloat_on_regain	on	network_signal_integrity_refloat_on_regain
network.signal_integrity.resume_warmup_vet	on	network_signal_integrity_resume_warmup_vet
network.signal_integrity.withhold_instead	off	network_signal_integrity_withhold_instead
network.signal_roster	off	network_signal_roster
network.sparse_station_normals	on	sparse_station_normals
network.state_snapshot_carry	off	state_snapshot_carry
network.state_snapshot_carry_diagonal	off	state_snapshot_carry_diagonal
network.state_snapshot_carry_lin	off	state_snapshot_carry_lin
network.state_snapshot_coupling_capture	off	state_snapshot_coupling_capture
network.state_snapshot_datum_capture	off	state_snapshot_datum_capture
network.state_snapshot_emit_capture	off	state_snapshot_emit_capture
network.state_snapshot_frame_capture	off	state_snapshot_frame_capture
network.state_snapshot_keep_history	off	state_snapshot_keep_history
network.state_snapshot_require	off	state_snapshot_require
network.state_snapshot_rxcb_lin_capture	off	state_snapshot_rxcb_lin_capture
network.state_snapshot_signal_capture	off	state_snapshot_signal_capture
network.state_snapshot_station_capture	off	state_snapshot_station_capture
network.station_marker_to_arp	off	network_station_marker_to_arp
network.station_pd_pin_aware	off	network_station_pd_pin_aware
network.station_prior_before_pd	off	station_prior_before_pd
network.sticky_pivot	off	network_sticky_pivot
network.streaming.msm_lli_from_locktime	off	network_streaming_msm_lli_from_locktime
network.streaming.require_datum_station	on	network_streaming_require_datum_station
network.trace_amb_if	off	trace_amb_if
network.trace_amb_state	off	trace_amb_state
network.trace_anchor_info	off	trace_anchor_info
network.trace_clock_full	off	trace_clock_full
network.trace_consumer_sat	off	trace_consumer_sat

JSON path	Default	Legacy alias
network.trace_dd_fractional	off	trace_dd_fractional
network.trace_epoch_status	off	trace_epoch_status
network.trace_estimability	off	trace_estimability
network.trace_estimability_deficit_only	off	trace_estimability_deficit_only
network.trace_events	off	trace_events
network.trace_flag_rates	off	trace_flag_rates
network.trace_ionosphere	off	trace_ionosphere
network.trace_phase_anchor	off	trace_phase_anchor
network.trace_phase_colour	off	trace_phase_colour
network.trace_postfit_keyed	off	trace_postfit_keyed
network.trace_postfit_rows	off	trace_postfit_rows
network.trace_postfit_sigma0	off	trace_postfit_sigma0
network.trace_prefit_census	off	trace_prefit_census
network.trace_raim_redundancy	off	trace_raim_redundancy
network.trace_rx_states	off	trace_rx_states
network.trace_sat_bias_states	off	trace_sat_bias_states
network.trace_section_timers	off	trace_section_timers
network.trace_sh_spectrum	off	trace_sh_spectrum
network.trace_slip_events	off	trace_slip_events
network.trace_slip_raw	off	trace_slip_raw
network.trace_ssr_consumer	off	trace_ssr_consumer
network.trace_station_block	off	trace_station_block
network.trace_station_pd_null	off	trace_station_pd_null
network.trace_station_residuals	off	trace_station_residuals
network.trace_stations	on	trace_stations
network.trace_topology	off	trace_topology
network.trace_zwd	off	trace_zwd
network.ura_retro_bump	off	network_ura_retro_bump
network.use_sif_coordinates	off	use_sif_coordinates
network.write_run_manifest	off	write_run_manifest
network.zwd_seam_reseed	off	zwd_seam_reseed
processing.amb_level_check	off	amb_level_check
processing.ar_ignore_mw_wl	off	ar_ignore_mw_wl
processing.auto_report	off	auto_report
processing.canonical_signal_code_follows_phase	on	canonical_signal_code_follows_phase
processing.enable_neq_export	off	enable_neq_export
processing.enable_noncausal_slip_bridge	off	enable_noncausal_slip_bridge
processing.enable_noon_turn_attitude	off	enable_noon_turn_attitude
processing.enable_obs_skip_to_latest	off	enable_obs_skip_to_latest
processing.enable_obs_ssr_sync	off	enable_obs_ssr_sync
processing.enable_orbit_normal_attitude	off	enable_orbit_normal_attitude
processing.enable_ssr_bias_max_age	off	enable_ssr_bias_max_age

JSON path	Default	Legacy alias
processing.enable_station_info	off	enable_station_info
processing.enable_trig_sp3_interp	off	enable_trig_sp3_interp
processing.ephemeris_jump_reseed	off	ephemeris_jump_reseed
processing.exclude_eclipsing_satellites	off	exclude_eclipsing_satellites
processing.inhouse_bias_estimation	off	inhouse_bias_estimation
processing.innovation_screen	off	innovation_screen
processing.network_consumer.bias	on	network_consumer_bias
processing.network_consumer.clock	on	network_consumer_clock
processing.network_consumer.enable	off	network_consumer_enable
processing.network_consumer.undifferenced	on	network_consumer_undifferenced
processing.relinearize_geometry	off	relinearize_geometry
processing.report_async	off	report_async
processing.rinex_skip_zero_values	off	rinex_skip_zero_values
processing.run_smoother	off	run_smoother
processing.ssr_bias_band_fallback	off	ssr_bias_band_fallback
processing.ssr_level_step_inflate	off	ssr_level_step_inflate
processing.ssr_pb_hold_presence_flip	off	ssr_pb_hold_presence_flip
processing.ssr_phase_bias_disc_carry	off	ssr_phase_bias_disc_carry
processing.ssr_phase_bias_step_translate	off	ssr_phase_bias_step_translate
processing.uncombined_enabled	off	uncombined_enabled
receiver.enable_antenna_orientation	off	enable_antenna_orientation
receiver.enable_antenna_substitution	off	enable_antenna_substitution
receiver.enable_itrf_velocity	off	enable_itrf_velocity

1.6 Model-correction switches

Table 6: Physical model on/off switches (ModelCorrections). These gate the correction terms applied to the undifferenced observation model.

Field	Default	Effect
enable_troposphere	on	Saastamoinen ZHD + GMF dry mapping (slant hydrostatic delay).

Field	Default	Effect
enable_klobuchar_ionosphere	off	Klobuchar ionospheric delay model (single-frequency fallback only). IF combinations and DD short-baselines cancel ionosphere geometrically; PPP-RTK uncombined estimates per-satellite slant iono states. In all three of those configs the Klobuchar model is a no-op, but constructing the Ionosphere instance still pays for an extra Model::Geometry load and the Klobuchar coefficient parse. Default off; intended only for single-frequency PPP runs that need an a-priori ionosphere.
enable_higher_order_ionosphere	off	Apply second-order ($B \cdot \cos(\theta) \cdot STEC/f^3$) and third-order ($\eta \cdot STEC^2/f^4$) ionospheric delays as a per-signal observation-level correction, evaluated at the GIM-derived IPP using a centered tilted dipole geomagnetic field. Requires a loaded IONEX GIM as the absolute-STEC source; the engine auto-loads the GIM when this flag is on even if estimation.ionosphere_weighting.enabled is false. No-op when no GIM resolves (warning logged at startup). Magnitudes: I_2 ˜1–2 cm at L1 during high TEC, I_3 ˜mm; the IF combination only partially cancels these, so the largest visible benefit is in standard PPP (IF). For PPP-RTK uncombined and short-baseline DD the higher-order term mostly cancels in the SD / DD differencing, so this flag is closer to a consistency knob than a bias remover. Default off.
enable_relativistic_correction	on	Periodic relativistic clock correction (-2 r·v / c^2).
enable_shapiro_delay	on	Shapiro (gravitational path) delay.
enable_solid_earth_tide	on	IERS 2010 solid Earth tide displacement.
enable_ocean_tide_loading	on	Ocean tidal loading displacement (BLQ).
otl_from_sif	off	Take the ocean tidal loading coefficients from the Station Information File registry (three components, FES2014b) without arming processing.enable_station_info, which would also replace the a-priori position and the ocean pole tide source. The registry is loaded from files.station_info_path(s) or ¡data¿/static/sif/. A station the registry does not resolve falls back to BLQ, then EOT20 (radial only), then zero, exactly as before. The SIF east and north columns are read as the BLQ west and south rows (the M-PAGES reading, checked against HARDISP), not as true east and north. Only used when enable_ocean_tide_loading is on. Default off keeps runs byte-identical.

Field	Default	Effect
otl_nodal_corrections	off	Apply the 18.6-year nodal amplitude factors f and phase corrections u to the 11 loading constituents, and take the Greenwich argument from UTC instead of GPS time (a 0.15 degree shift on M2). Sub-millimetre to about 1 mm at ocean sites; the IERS HARDISP reference gets the same effect from its admittance side lines. Only used when enable_ocean_tide_loading is on. Default off keeps runs byte-identical.
otl_remove_diurnal_translation	off	Remove the net translation (a uniform ECEF displacement) of the SIF diurnal loading field, K1 O1 P1 Q1, from the coefficients of each station served by otl_from_sif. The SIF loading is relative to the centre of mass; on consumer replays of CODE-based network products the translation is not seen in the data (north-component scale -0.11, 95 % interval -0.75 to +0.27, against 1.19 for the rest of the model, 13 sites in 11 six-hour windows), and it carries 5.7 of the 8.0 mm of north loading at WTZZ00DEU in the 01 to 06 UT window of 2024-129. The table holds for the IGS20 SIF registry (scripts/otl_compare/net_translation.py). Only used with otl_from_sif and enable_ocean_tide_loading. Default off keeps runs byte-identical.
enable_pole_tide	on	Pole tide displacement (from ERP): IERS 2010 Sec. 7.1.4 rotational deformation.
enable_ocean_pole_tide	off	Ocean pole tide load displacement (IERS 2010 Sec. 7.1.5 / Desai 2002), added on top of enable_pole_tide. Requires the opole load grid (files.opt_grid_path or ¡data¿/static/opole/). ˜0.1-2 mm; default off keeps runs byte-identical.
enable_atmospheric_loading	off	Atmospheric pressure loading (S1/S2 thermal-tide site displacement). Requires per-station S1/S2 load coefficients (files.atm_tide_grid_path or ¡data¿/static/apl/). Sun-synchronous, so it does not average out in static daily PPP. Default off keeps runs byte-identical.
enable_satellite_pco	on	Satellite antenna phase centre offset.
enable_satellite_pcv	on	Satellite antenna phase centre variation.
enable_receiver_pco	on	Receiver antenna phase centre offset.
enable_receiver_pcv	on	Receiver antenna phase centre variation.
enable_receiver_cross_system_atx_fallback	off	Receiver only: when the receiver ATX entry lacks a non-GPS frequency, substitute the nearest GPS band (E1-¿G01, E5a-¿G05, B1C-¿G01, B2a-¿G05, ...). Valid because the receiver phase pattern is frequency-dependent on shared hardware. Never applied to the satellite antenna (E01 and G01 are different spacecraft). Opt-in.

Field	Default	Effect
enable_atx_frequency_fallback	on	When the ATX entry does not have PCO/PCV for the requested frequency (common for L5 / E5* / B2a on older satellites or receiver antennas), look up the nearest available frequency on the same antenna and use its values instead of skipping the correction without notice. Cross-frequency PCO drift on a single physical antenna is typically ¡1 cm, better than the few-cm vertical residual that comes from skipping a frequency entirely. false restores the legacy "skip if missing" behaviour for A/B comparison.
require_atx_coverage	off	When true, drop observations whose (sat, signal) lacks complete ATX coverage on both the satellite and receiver entries, i.e., satellite PCO+PCV and receiver PCO+PCV must all resolve to non-empty entries for the row's exact frequency, without enable_atx_frequency_fallback substituting a neighbouring frequency. Default false (legacy behaviour: observations are kept and any missing PCO/PCV is zeroed without notice). True suits a run with 'frequency_mode: "all"' (legacy alias 'signal_mode') in which the ATX file serves as the authoritative whitelist, dropping rows that would otherwise be processed without proper antenna calibration.
require_atx_coverage_for_ar	off	When true, AR (WL constraints, LAMBDA NL search) skips (sat, signal) pairs lacking strict ATX coverage even though the underlying observations are still used in the float Kalman update. Softer counterpart to require_atx_coverage: keeps "all"-signal float estimation while restricting integer fixing to the ATX-curated subset. Default false (no AR-side gating).
restrict_ar_for_substituted_atx	off	When true, AR additionally skips (sat, signal) pairs whose ATX correction was computed via the frequency-fallback substitution chain (rather than an exact entry). Targets specifically the cases where the antenna model is approximate; signals lacking ATX entirely (no substitution applied) are still allowed. Usable alongside or instead of require_atx_coverage_for_ar, depending on which kind of imperfection is to be gated. Default false.
loosen_amb_q_for_substituted_atx	off	When true, ambiguity states whose underlying (sat, signal) had their ATX correction frequency-fallback-substituted use a looser process noise (ambiguity_q_substituted_atx) instead of the standard process_noise.ambiguity. Compensates for the small but non-zero modeling error introduced by the substituted PCV/PCO. Default false.

Field	Default	Effect
apply_phase_shifts	off	When true, RINEX phase observations have the receiver-applied "SYS / PHASE SHIFT" values subtracted at parse time, recovering the unshifted phase convention. Default false because most modern BIA products (CODE, CAS, GFZ) are calibrated against the receiver-shifted phases as recorded in RINEX, so leaving the shifts in place is the standard PPP convention. True only when the bias product is explicitly calibrated for the unshifted convention, or when running without external BIA and using in-house bias estimation that benefits from a uniform shift-free input.
enable_phase_windup	on	Carrier-phase wind-up correction.
enable_code_phase_bias	on	Code/phase bias corrections from BIA product files.
bias_coverage_requires_apply	off	When true, the "already corrected?" gates in
refine_emission_time	on	When true, Geometry::computeSatelliteGeometry runs a

1.7 Logging and trace channels

Table 7: Logging and CSV-trace channels (LoggingConfig). Each toggles one spdlog file logger or one per-epoch CSV trace.

Field	Default	Channel
enable_main_log	on	logs/main_log.log
enable_gnss_config	on	logs/gnss_config.log
enable_kalman_filter	on	logs/kalman_filter.log
enable_smoother	on	logs/smoother.log
enable_state_manager	on	logs/state_manager.log
enable_ppp_model	on	logs/ppp_model.log
enable_measurement_model	on	logs/measurement_model.log
enable_mw_slip_detector	on	logs/mw_slip_detector.log
enable_if_slip_detector	on	logs/if_slip_detector.log
enable_gf_slip_detector	on	logs/gf_slip_detector.log
enable_turbo_edit_resolver	on	logs/turbo_edit_resolver.log
enable_cycle_slip_manager	on	logs/cycle_slip_manager.log
enable_ambiguity_resolver	on	logs/ambiguity_resolver.log
enable_corrections	on	logs/corrections.log

Field	Default	Channel
enable_geometry	on	logs/geometry.log
enable_troposphere	on	logs/troposphere.log
enable_ionosphere	on	logs/ionosphere.log
enable_antenna	on	logs/antenna.log
enable_antenna_model	on	logs/antenna_model.log
enable_ocean_tide	on	logs/ocean_tide.log
enable_planetary_model	on	logs/planetary_model.log
enable_stochastic_model	on	logs/stochastic_model.log
enable_rinex_parser	on	logs/rinex_parser.log
enable_product_parser	on	logs/product_parser.log
enable_product_cache	on	logs/product_cache.log
enable_resolver	on	logs/resolver.log
enable_state_initializer	on	logs/state_initializer.log
enable_bssd_pivot_manager	on	logs/bssd_pivot_manager.log
enable_bssd_transform	on	logs/bssd_transform.log
enable_ppp_utils	on	logs/ppp_utils.log
enable_downloader	on	logs/downloader.log
enable_ntrip	on	logs/ntrip.log
enable_rt_engine	on	logs/rt_engine.log
enable_imu	on	logs/imu.log
enable_state_trace	on	logs/state_trace.csv
enable_ambiguity_trace	on	logs/ambiguity_trace.csv
enable_ionosphere_trace	on	logs/ionosphere_trace.csv
enable_sat_bias_trace	on	logs/sat_bias_trace.csv
enable_code_bias_trace	on	logs/code_bias_trace.csv (receiver DCB/IFB)
enable_sat_status_trace	on	logs/satellite_status_trace.csv
enable_dop_trace	on	logs/dop_trace.csv
enable_residual_trace	on	logs/measurement_residuals.csv
enable_wls_trace	on	logs/wls_trace.csv
enable_smoother_trace	on	logs/smoother_trace.csv
enable_mw_slip_trace	on	logs/mwwl_trace.csv
enable_if_slip_trace	on	logs/if_slip_trace.csv
enable_gf_slip_trace	on	logs/gf_slip_trace.csv
enable_turbo_edit_trace	on	logs/turbo_edit_trace.csv
enable_windup_trace	off	logs/windup_trace.csv (phase wind-up diagnostic; off by default)
kalman_filter_concise_log	off	

1.8 Cycle-slip detector registry

Table 8: Cycle-slip detectors known to `SlipDetectorRegistry`. *Active* detectors have a creator in the factory table and run when their flag is set; *staged* detectors are declared in the `SlipDetectorKind` enum but excluded from the build (the registry warns and skips them if requested). The $k\sigma$ column is the MAD gate multiplier from `CycleSlipConfig`.

Detector	Config flag	Default	$k\sigma$	Status	Combination
`MelbourneWubbena`	`estimation.cycle_slip.mw`	on	8.0	**active**	Wide-lane MW combination (primary; the registry always has a creator for it).
`Graphic`	`estimation.cycle_slip.graphic`	off	6.0	**active**	GRAPHIC (code+phase average) per-signal detector.
`ExtraWideLane`	`estimation.cycle_slip.ewl`	off	5.0	**active**	Extra-wide-lane MW for satellites with 3 or more frequencies.
`GeometryFree`	`estimation.cycle_slip.gf`	off	–	*staged*	2nd-order time-differenced geometry-free (staged: not compiled, no registry creator).
`IonoFreePhase`	`estimation.cycle_slip.if`	off	–	*staged*	Between-satellite, between-epoch IF phase OMC; no registry creator (network.cycle_slip.if_phase_td).

1.9 State-vector key schema

Table 9: Dynamic state-vector key formats, reconstructed from the `make*Key` bodies in `StateKeySchema.hpp`. Angle-bracketed fields are runtime values; literal characters are the delimiter / infix / prefix / suffix constants.

State	Key format	Builder
Ambiguity	`<satellite>_<signal>`	`makeAmbiguityKey`

State	Key format	Builder
NL Ambiguity	<satellite>_NL_<band>	makeNlAmbiguityKey
WL Ambiguity	<satellite>_WL_<band_a>_<band_b>	makeWlAmbiguityKey
DD Ambiguity	DD|<cons>|<sat_i>|<code>|<pivot>	makeDdAmbiguityKey
Ionosphere	<satellite>	makeIonosphereKey
IFB	<system>_IFB<band>	makeIfbKey
DCB	<system>_DCB	makeDcbKey
Glonass Per Slot IFB	<system>_IFB<band>_k	makeGlonassPerSlotIfbKey
Receiver Phase Bias	<system>_RPB_<signal>	makeReceiverPhaseBiasKey
Receiver Isc	<system>_ISC_<signal>	makeReceiverIscKey
Satellite Bias	<satellite>|<signal>	makeSatBiasKey
Satellite Phase Bias	<satellite>|<signal>|PB	makeSatPhaseBiasKey
Satellite Clock	<satellite>|SC	makeSatClockKey
System Time Offset	<system>	makeSystemTimeOffsetKey
Pivot Seed	<cons>|<band>	makePivotSeedKey
Vtec Sh	|||<trig>	makeVtecShKey